Assignment 1

Assignment: Build a Semantic Search Engine over PDF Documents

Objective

Build a semantic search system that allows users to search inside a collection of documents using natural language queries.

The system should retrieve the most relevant document sections based on semantic similarity instead of keyword matching.

Requirements

You are given a folder containing multiple PDF documents.

Your task is to:

1. **Load Documents**

Read multiple PDF files.

Extract their text.

2. **Implement Text Chunking**

Create a function:

that: Splits documents into smaller chunks.

3. **Generate Text Embeddings**

Use: SentenceTransformer

to convert every chunk into a vector representation.

Example:

4. **Build FAISS Vector Database**

Students should:

Normalize embeddings.

Create FAISS index.

Store all document vectors.

5. **Implement Search Function**

Create:

semantic_search(query, top_k)

The function should:

Convert query into embedding.

Search FAISS.

Return top matching chunks.

6. **Test the System**

Example:

Query:

How many vacation days do employees receive?,

Expected:

Employee_Handbook.pdf

Relevant chunk: -> " ..."

Test Cases

Query,	Expected PDF
How many vacation days do employees receive?, ->	Employee_Handbook.pdf
What is the refund policy?, ->	Customer_Service_Policy.pdf

How is customer satisfaction improved?, -> Customer_Service_Policy.pdf

What metrics evaluate classification models?, -> Machine_Learning_Basics.pdf

Explain transformer architecture., -> Deep_Learning_Guide.pdf